Thanos Vassilakis

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Distinguished Engineer, Managing Director Head of AI and Architecture for Finance IT, RBC Capital Markets (Sep 2025 – Present).

EXECUTIVE SUMMARY

Accomplished technology executive with over 20 years of experience driving transformation and innovation in the financial services sector. Demonstrated expertise in leading large-scale development initiatives, leveraging disruptive technologies, and fostering an entrepreneurial culture within established organizations. Proven record of enhancing income generation through strategic technology integration and operational efficiency. Adept at creating inclusive environments that empower teams and encourage collaboration to drive progress and maximize business impact. My career is characterized by a commitment to continuous learning, mentorship, and the pursuit of excellence in technology and its application in the finance industry.

CURRENT MANDATE

- Technical ownership of GL migration from zOS/DB2 to Linux/Oracle and Sub-ledger modernization.
- Drive modernization and AI adoption of RBC Finance IT.

WHAT I'VE RUN AT RBC

A multi national group of 166, 47% diversity, and with full-time teams in Canada, US, UK, Malaysia, Mexico, and consultants in India with a current budget of just over 30m.

- **AI solutions and modernization for the General Ledger and ERP.**
- **Architecture for Operations** -- driving modernizing in reporting, monitoring, and the confirmation and settlement landscape, plus overseeing all development and vendor acquisitions in operations.
- **Access Control** -- a central access management services for 1k+ apps and services in Capital Markets.
- **Enterprise Reconciliation Service** -- built in-house processing 300 realtime and 2k+ daily reconciliations using ML and GPU matching.
- **Trade Expense Management.**
- **Expense Reporting Service** for Capital Markets.
- **Global Market Supervision** -- an event driven service that is now using ML and hierarchical agents state service to model behavior and generate required reports by SEC, FINRA, UK and European regulators.
- **Market Access, Trade Limits Control** -- a Rule 15c3-5 compliant centralized ML automated trade limits for most all our order management systems.
- **ML Pricing Service** -- used by the Municipal Bond and Credit Derivative desks. Predicts a fair price for all unique 1.5m bonds in real-time and has beaten our competitor evaluation services every single trading day since it went live 4 years ago.
- **All Shared Operation Services** -- UI Platforms, Reporting Service, central queuing, Notification, FAS, T+1 Reporting, + Trade Repository for Capital Markets, Wealth Management and Cash Management.
- **R&D for Technology** -- blockchain, crypto, ML model development, plus inner source and open source stewardship.

KEY ACHIEVEMENTS & DELIVERABLES

AI, Machine Learning & Quantitative Platforms

- Designed the Strategy for Data for AI.
- Modernization of the GL and Posting Engines with AI.
- **Pricing & Risk:** Developed ML-based real-time pricing models for municipal bonds and credit derivatives, alongside real-time Position and PnL systems (2020).
- **ML Infrastructure:** Delivered an ML Model execution and scoring service (2018) to operationalize data science initiatives.
- **Data Strategy:** Established the Capital Markets Data Lake (2016) and a Single Data Interface SDK (2019) to standardize data access.

Strategic Cost Optimization & Vendor Consolidation

- **Enterprise Reconciliation Service (2025):** Architected an in-house solution using ML and GPU matching to displace legacy vendor products, securing **$7M in annual savings**.
- **Trade Expense Services (2024):** Deployed agentic invoice processing to modernize expense management, eliminating external vendor dependencies for **$11M/year savings**.
- **Global Market Supervision (2023):** Engineered an event-driven, real-time supervision service that replaced third-party systems.
- **Post-Trade Modernization (2024):** Built T+1 infrastructure and post-trade processing services, ensuring regulatory compliance.
- **Access Control Modernization (2023):** Centralized application access control for over 1,000 apps.
- **Governance & Data (2023):** Internalized Reference Data Services and ML-based Trade Limits Governance.

Core Infrastructure & Technology Transformation

- **Engineering Leadership:** Built and scaled high-performance development teams across Kuala Lumpur, Halifax, and Montreal.
- **Language Adoption:** Successfully introduced and standardized Python (2010) and Elixir (2017) within the enterprise technology stack.
- **Platform Engineering:** Architected critical infrastructure including Distributed Compute Service (2018), Function-as-a-Service (Lambda) (2017), and Central Logging (2017).
- **Trading Cores:** Built the Real-time Trading Data Core (Athena-style, 2018) and the Cash Management Real-time Core with virtual accounts.
- **Innovation:** Developed a Tokenization Platform (2022) to advance digital asset capabilities.

Regulatory & Compliance

- **Risk & Reporting:** Delivered infrastructure for FRTB (Fundamental Review of the Trading Book) (2018) and Anti-Terrorism/Money Laundering services (2018).
- **Exchange Migrations:** Led the LME trading platform development and migrated commodity trades onto OpenLink.

WORK EXPERIENCE

RBC Capital Markets | New York

- Distinguished Engineer, Managing Director, Head of AI and Architecture for Finance IT — Sep 2025 – Present
- Managing Director, Distinguished Engineer, Operations Infrastructure — Sep 2023 – Sep 2025
- Distinguished Engineer, Head of Technology — Jan 2022 – Sep 2023
- Head of Shared Trading Services, Head of R&D for Technology & Head of Reporting Services — Nov 2016 – Jan 2022
- Head of R&D for Technology — Nov 2014 – Jan 2016

Infinigon Group Inc | Head of Development | 2012 – 2014

- Created a cross-exchange custodial cryptocurrency trading platform.
- Built a machine learning analytics platform utilizing real-time data and social media sentiment.

RBC Capital Markets | Consultant | Sep 2010 – Jan 2012

- Built commodities platform for oil, gas and LME.

Credit Suisse First Boston | Consultant | Oct 2007 – Sep 2010

- Architected a risk system for complex commodities trades and integrated external pricing models for OpenLink.
- Built a central trade repository and market data services for commodities.

JPMorganChase | Consultant | Feb 2007 – Nov 2007

- Contributed to the Athena platform, developing a code profiler and native FIX engine.
- Developed a standardized trade feed framework and implemented gateways for CME and ICE to streamline energy trading operations.
- Optimized network and computation performance for the counterparty risk system (Gauss).

Zingy | Director of Mobile Applications | Jan 2006 – Feb 2007

- Managed development and operations for 50+ mobile applications across 27 carriers and multiple platforms.
- Led the development of innovative mobile products, including the top-selling MapQuest mobile application.

ICAP | Director of Front-End Development | Feb 2004 – Jan 2006

- Established and led the in-house development team for fixed income trading systems.
- Developed high-performance distributed trading services and mobile trading platforms.

NYSE / SIAC | Technical Director / Display Book | Oct 2000 – Oct 2004

- Architected and implemented market surveillance systems, stock information servers, and regulatory reporting services.
- Developed innovative Python-based trading services for specialists.

BIO

I came to Computer Science by accident. At eighteen, I had no interest in computers; my heart was set on being a hydrologist to solve water problems in developing countries. My technical interests were analog -- trading and repairing vintage valve amplifiers. That changed during a summer job at the Electric Coin Company, a slot machine vendor. I expected to wield a soldering iron but found myself coding 6801 controllers. By the end of the summer, I was addicted to programming. I switched tracks, graduating from North London Polytechnic with a degree in Maths & Computer Science.

My early career was defined by eclectic, high-intensity technical challenges. I worked with Richard Forsyth on *Darwin*, one of the earliest Machine Learning frameworks, which we used to unsuccessfully predict horse races. I worked in Atari's PC games division, built RIP engines for printers, and developed a real-time OS for Ferranti's Lynx helicopter program. My freelance work spanned industries: fuzzy logic control systems for Unilever, video routing for the BBC, and neural networks on Transputers for the UK Treasury.

In 1993, while working at Apple on the Newton program, I discovered the internet and built a WAIS-based patient record system for the UK National Health Service. A pivotal moment came in 1995 when I was commissioned to build the first internet casino for Caesars. My initial C++ codebase was tight but unwieldy at over 40,000 lines, and I struggled to complete it. On a whim, I bought Mark Lutz's Python book and rewrote the entire system over a weekend. The result was a superior, 4,000-line solution. That project made me a lifelong Python evangelist and led to the creation of an early cyber currency, WebCheck.

After stints in Europe -- advising the European Parliament, building Erlang systems for Ericsson, and being poached by IBM Research in Zurich to code a mobile JVM -- I returned to my hometown, New York, in 1997. I architected the NYSE's first internet-based system, which was likely the first financial production system written in Python. Since then, my focus has been on high-frequency trading, fixed income platforms, and risk systems for major institutions like ICAP, JP Morgan, Credit Suisse, and RBC Capital Markets. Between these corporate mandates, I continue to explore the bleeding edge through running a startup style group within the bank.

SOME KEY ACHIEVEMENTS & INDUSTRY FIRSTS

Achievement	Description
Europe's 1st Missing Children's Registry	Built a central online database of missing children and was awarded a citation from the European Parliament.
European Parliament workflow service	Handles the complete workflow of all artifacts in the European Parliament
Top earning US mobile apps	including MapQuest, eBay, NBC Weather, New York Times
Phonescript	1st phone programming language
1st Python trading service	Built for SIAC/NYSE
ICAP's fixed income trading platform	Still used by most banks for fixed income trading.
1st C++ debugger & Numeric library	Coded standard libraries for Walter Bright's C++ distribution
1st Internet Casino	Caesars Casino
1st Web framework for Python (PSO)	Included in Debian Linux and used by European Parliament, NASA, SIAC, JP Morgan
Byteeasy	Anonymous crypto exchange
BTC Terminal	Cross-exchange custodial crypto trading platform
JVM for Mobile	Coded the mobile phone version of the JVM for IBM
Author of Nimalana	Nim based Solana validator
Developer of Aegis	A Zig based high-performance Bittensor miner specifically engineered to exploit Apple Silicon

DOMAINS & COMPANIES

Domain	Companies
Trading & Risk	Infinigon Group, RBC CM
Blockchain & Crypto	RBC CM, Infinigon Group, Solana, Marita Bank, Caesars Casino, Electric Coin Co.
Machine Learning & Big Data	RBC CM, Infinigon Group, Caesars Casino, Ericsson
Mobile Development	Apple, RBC CM, IBM, ICAP, Ericsson, Zingy
Gaming & Robotics	Atari, Zingy, Caesars Casino, Unilever
High Performance Computing	JP Morgan, NYSE, SIAC, HP
Telecommunications	British Telecom, Ericsson, HP, Honeywell Bull, Multitone, Reuters
Cloud & Distributed Systems	RBC CM, Infinigon Group, Ericsson
Medical & Public Sector	British National Health, European Parliament, European Community, AAH Meditel
Military	Ferranti

STACK OF PAST WORK

RBC CM, Infinigon Group, RBC CM, JP Morgan, Credit Suisse, JP Morgan, Zingy, ICAP, SIAC/NYSE, Ericsson/HP, West LB, European Community, Caesars Casino, European Parliament, AAH Medical, Apple, Symantec (Zortech), Reuters International, Honeywell Bull, Drake, British Telecom, Ferranti, Barclays, Multitone, Atari, Unilever, Atari, Electric Coin Company.

LANGUAGES

C, C++, Erlang, Elixir, Fortran, Go, Java, JavaScript, Python, Solidity, Zig, some Rust, COBOL.

PERSONAL DETAILS

- **Nationality:** US & Greek (European)
- **Spoken Languages:** English & French
- **Email:** thanosv@gmail.com

WORK REFERENCES

- Kabir Reike, MD Head of Data Architecture, Bank of America
- Sid Belzberg, CFO, Infinigon Group - ex Captain of the Canadian Chess Team
- Endri Mani, MD Head of GenAI, RBC CM
- Yarik Chinskiy, Head of Data, AI and Corporate Tech, TD
- Geraud Boyer, Director of Engineering, Datadog

PERSONAL REFERENCES

- Bruno Dupire, Head of Quantitative Research, Bloomberg
- Hope Igelhart, Senior Psychoanalyst, Columbia University
- Stavros Lambrinidis, European Union ambassador to the UN